

Socially Assistive Robots as Decision Makers: Transparency, Motivations, and Intentions

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ABSTRACT

Socially Assistive Robots (SARs) are being developed to fulfil a range of roles that support humans. As the complexity and capability of SARs increase, they will be expected to adopt higher degrees of responsibility and execute greater levels of autonomous decision-making. Therefore, it is imperative that the Human-Robot Interaction (HRI) and more widely the Human-Computer Interaction (HCI) community seriously consider how SARs communicate about their role and the motivations and intentions behind their decisions. The proposed workshop will address challenges with respect to SAR decision-making, discuss current approaches to these challenges, and develop ideas and strategies for how the wider CHI community should move forward in this area.

CCS CONCEPTS

• **Computer systems organization** → **Robotic autonomy**; • **Security and privacy**; • **Human-centered computing** → **Accessibility**;

KEYWORDS

Socially assistive robots, Transparency, Trust, Robotic autonomy, Machine ethics

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1 MOTIVATION

Socially Assistive Robots (SARs) are becoming increasingly relevant in the human-robot interaction space with promising applications in healthcare, education, assistive living, physical coaching, and beyond. They can allow for more seamless integration into the existing social dynamic of the environment through intelligent means, as opposed to forcing the user to adapt to the limitations of some traditional technology intervention [3]. Furthermore, SARs' physically embodied have been seen to drive greater engagement over the likes of virtual agents in assistive scenarios, for example, where they have been deployed as exercise coaches [2] [5]. Beyond the benefits for the end-user, real-world SAR interventions also have the potential to support assistive staff, through monitoring, reporting and task engagement. For the robot to reach this level of autonomy, greater decision-making responsibilities will need to be handed over. This shift towards SARs as key decision-makers creates new challenges for the HRI community that we aim to shed light on during this workshop.

One differentiating factor of social robots is their exposure to users who are not knowledgeable of their motivations or intentions, something that is not necessarily obvious at face value. In addition

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to this, in human-human interactions, assistive staff are considered experts in their area of practice and are therefore trusted in their decisions, or otherwise when prompted can reason as to why decisions were made. In scenarios where the SAR makes key decisions, an important consideration will be how to build transparent and trustworthy interactions in which the robot can communicate not only its motivations and intentions but justification and reasoning around its individual choices. Such transparency is becoming an ever-growing challenge as personalization and adaption are now recognized as key factors in achieving greater engagement in these systems [6]. Because of this, research is progressing from simple rule-based decision systems to complex decision-making mechanisms such as machine learning approaches. These techniques have provided promising results, however, they are notoriously difficult even for humans to explain to one another.

To complicate this further, research has shown that human trust can be significantly impacted by errors during HRI [4]. Providing the robot can detect errors in its decision-making in the first instance, effectively handling such errors to retain user trust will be an important consideration in the design of such interactions. Moreover, SARs are typically seen to operate within sensitive populations and on challenging tasks. Although frameworks have been introduced in recent years to consider ethical and cultural differences when designing these systems [1], other characteristics of the interaction such as physical or cognitive impairments create new considerations for how these agents should communicate their decisions.

2 CHALLENGES TO ADDRESS

There are various challenges that arise as SARs increasingly take on decision-making responsibilities that historically would have been in the hands of humans. Challenges are exacerbated particularly in interactive scenarios where the end-user is vulnerable and/or has some form of physical/cognitive impairment. The objective of this workshop is to bring together academic and industrial experts in the areas of HCI and HRI to discuss the following challenges:

- Determining **which decisions should be made by the robot** can be difficult. Where do we draw the line, and who decides this?
- How can SARs **transparently communicate their decision-making to the user when underlying reasoning is based on complex processes** that are not always straightforwardly interpretable or explainable (i.e., black-box processes)?
- How do SARs smoothly integrate into existing relationships across different scenarios to **build trust** with users?
- What is the **role of robot embodiment** in communicating explanations (which features can be more effective and when)?
- How do SARs **recover from unintended/incorrect actions** through communication with the user and ensure this does not negatively affect trust? That is, assuming the SAR learns via some mechanism that its previous action(s) were incorrect.

- How should SARs' approach and means of interaction be **adapted for specific populations** (e.g., informed consent for individuals with cognitive impairments)?
- Should SARs attempt to **implement a normative moral framework** together with autonomous decision-making?

3 ORGANIZERS

Emilyann Nault, University of Edinburgh (UK) Main Point of Contact. Emilyann Nault is a fourth-year PhD student researching how socially assistive robots and sensory feedback can be used to foster engagement with cognitive activities for older adults. She has integrated Participatory Design and user-centered design methodologies to engage end-users and relevant stakeholders throughout the research process.

Carl Bettosi, Heriot-Watt University (UK). Carl Bettosi is a second-year PhD student whose research focuses on adaptive socially assistive robots for upper-limb rehabilitation. Specifically, Carl is employing machine learning techniques to help drive better engagement through more adaptive and personalized behavioral policies in the agent.

Lynne Baillie, Heriot-Watt University (UK) Secondary Point of Contact. Professor Lynne Baillie has been involved in user-centered design of robotic, home, mobile and rehabilitation technologies for over seventeen years. She has had several full papers published at the SIGCHI conference and has also previously run two workshops at CHI. She is currently the Director of the Interactive and Trustworthy Technologies Research Group at Heriot-Watt University. Her research work has focused on how we can design technology that better interacts with people in terms of convenience, usability, performance and security.

Ronnie Smith, University of Edinburgh (UK). Ronnie Smith is a PhD student researching how to bring humans 'in-the-loop' of their own assistive technology solutions, ultimately enabling pro-active robotic assistance during daily life. His research bridges themes in artificial intelligence and human-robot interaction, with recent work on activity recognition, active learning, conversational agents, and human-robot collaboration during activities of daily living.

Maja Matarić, University of Southern California (USA). Maja Matarić is the Chan Soon-Shiong Distinguished Professor of Computer Science, Neuroscience, and Pediatrics at the University of Southern California (USC), and founding director of the USC Robotics and Autonomous Systems Center. Her PhD and MS are from MIT, BS from Kansas University. She is Fellow of AAAS, IEEE, AAAI, and ACM, recipient of the Presidential Award for Excellence in Science, Mathematics & Engineering Mentoring, Anita Borg Institute Women of Vision for Innovation, NSF Career, MIT TR35 Innovation, and IEEE RAS Early Career Awards, and authored "The Robotics Primer" (MIT Press). A pioneer of the field of socially assistive robotics, her research is developing human-machine interaction methods for personalized support in convalescence, rehabilitation, training, and education for autism spectrum disorders, stroke, dementia, anxiety, and other major health and wellness challenges.

Vivek Nallur, University College Dublin (IRL). I work in the area of Machine Ethics. I am interested in how to implement and verify ethics in autonomous machines. Questions such as what kinds of ethics would autonomous machines agree to among themselves, how would we ensure that individually ethical machines don't combine to produce un-ethical behaviour, are interesting to pose and answer computationally. I was on the Organizing Committee for AAAI 2021 Spring Symposium Series on Implementing AI Ethics [22-24 March 2021]. I was also on the Program Committee for the 1st Computational Machine Ethics Workshop at KR2 2021. I am a full voting member, and serve on the IEEE P7008 Standards committee for Ethically Driven Nudging for Robotic, Intelligent and Autonomous Systems. Multi-Agent Systems (MAS) are my preferred tool for approaching such problems and simulating possible futures with different ethical frameworks.

Manfred Tscheligi, University of Salzburg (AUT). Professor Tscheligi works in the area of human-computer interaction, contextual interfaces and user experience research. This is related to a wide area of application contexts such as automation, production, energy, training, mobility or entertainment systems. He is (co)-author of several publications at relevant conferences, has (co)-organized a wide set of workshops (CHI, AUI, MobileHCI, Persuasive Technology, ECSCW, CSCW) and also co-chaired HRI 2107. He has initiated and managed a broad variety of national and international research as well industrial projects. Currently he is heading the Department for Artificial Intelligence and Human Interfaces at the University of Salzburg as well as the Center for Technology Experience at the Austrian Institute of Technology (AIT) in Vienna.

Andreas Sackl, Austrian Institute of Technology (AUT). is Scientist at AIT. He studied Media and Computer Science at the Technical University of Vienna and Mass Media and Communication Science at the University of Vienna, both finished with a Master degree. Furthermore, he holds a PhD in Computer Science from the Technical University of Berlin. He is author of numerous workshop and conference papers and acts as reviewer (TVX, ACM SIGCHI, etc.) and TPC member (QoMEX, PQS, QoE-FI). He is currently working in various national and international research projects covering UX, human-robot-interaction, acceptance modelling, accessibility, Quality of Experience and AAL.

Fabio Paternò, National Research Council of Italy - Institute of Information Science and Technologies (ITA). Professor Fabio Paternò is a Research Director at the CNR-ISTI. His main research interests are in Interactive Smart Spaces, Human-Robot Interaction, Accessibility, End-User Development, and Human-centered Artificial Intelligence.

Scott MacLeod, Heriot-Watt University (UK). Scott MacLeod is a PhD student whose research focus is using pervasive sensing, telepresence, and robotic technology, to facilitate remote assessment, and automate continuous cognitive assessment of people with mild cognitive impairment and/or at risk of developing dementia. To enable the provision of assistance always in tune with people's stages in life and care needs.

Sara Cooper, PAL Robotics (ESP). Sara Cooper (F) graduated in Biomedical Engineering at Mondragon Unibertsitatea (Spain) in

2018. In 2019 she obtained a Master of Science in Robotics at Heriot-Watt University with an AWE Prize for excellence in MSc project that was carried out as part of the Edinburgh Centre of Robotics, researching human-robot interaction in handover scenarios. She joined PAL Robotics in 2020 as a Robotics software engineer, working on healthcare-related European projects SHAPES and SPRING, where she is working with the social robot ARI for older care and hospital scenarios. Currently she is also involved in different software design of human-robot interaction such as chatbot design, perception and general behavior integration on the robot, as well as other pilots consisting in deploying robots at real life environments.

4 WEBSITE

The workshop website will be hosted on <https://sites.google.com/view/sar-decision-making>. It will contain information including, but not limited to, the workshop schedule, organizers' details, and the call for participation.

5 PRE-WORKSHOP PLANS

Interested parties will be invited to submit a short 4-page paper which engages with the overarching theme of decision-making around SARs. Prior to the workshop, a dedicated Discord server will be used to encourage discussion and begin building a sense of community. Discord has been chosen based on prior positive experiences with the platform, however, we will ask participants for their opinions and will adapt the platform if necessary to better serve the group. Participants will be asked to share sources of relevant research, demonstrations, news articles, etc. This will assist the organizers in determining how to best structure the discussion at the workshop in order to highlight the communities' interests and priorities. We will further reach out via Discord asking to let the organizers know privately (via Discord or email) if they have any accessibility requirements, which we will then organize through communication with the conference accessibility chairs.

6 HYBRID APPROACH

In order to include globally diverse participants, we plan to organize the workshop so participants may attend in person or online. To support this hybrid setup, we will question participants prior to the workshop so we can adapt our approach to include and support all in attendance. The synchronous discussion will be held in person and remotely via the Discord server. The asynchronous discussion will be facilitated through Discord. Certain organizers will be assigned the role of Support Chair to facilitate and assist remote participants.

7 ASYNCHRONOUS ENGAGEMENT

Resources will be provided before, during, and after the workshop in order to facilitate discussion and community building.

- (1) Before the workshop, the accepted papers will be accessible through the website. Also, there will be a dedicated Discord channel for participants to introduce themselves.
- (2) The outcomes of the workshop discussions will be produced and along with the online whiteboard tool (i.e., Miro) will be shared and used to virtually collaborate across synchronous and asynchronous participation.

- (3) After the workshop, relevant resources will be posted to Discord and the workshop website to encourage further asynchronous communication.

8 WORKSHOP STRUCTURE

The workshop will bring together experts in HRI and HCI from around the world who have encountered the challenges presented in this workshop around SAR decision-making. They will contribute their experiences, methodologies, and practices to build a holistic view of these challenges from a variety of disciplines. With this foundation, we will discuss conflicts within the ethical boundaries of these challenges, current approaches, and how they can be improved and better integrated going forward.

This will be a one-day workshop split into three sessions (Table 1). We expect 25-35 participants to be in attendance in person and will accept 15-25 short (4 pages) position or research papers for presentation. These presentations will take place on the day of the workshop and will vary in format (e.g., short talk, poster, demos, videos) in order to maximize engagement and collaboration between participants.

For the workshop activity, participants will be split into groups and each of which will receive a scenario in which a SAR acts as a decision-maker. They will be asked to design how the robot in their given scenario will make decisions and how they will be communicated to the user with respect to the challenges of the workshop. They can do so through written descriptions or sketches. Online participants will engage with the activity through the Miro boards and Discord channel, and an organizer will be available to help facilitate. The goal of this activity is to develop concrete strategies towards SAR decision-making regarding what information and how it is communicated to the user. It will also provide a practical means to discuss these overarching challenges. The final session of the workshop will consist of an open discussion of themes that have been identified throughout the day and how we can move forward in the space of SAR decision-making. The proposed schedule is as follows:

Table 1: Workshop Schedule

9:00 - 9:30	Introduction & Welcome
9:30 - 12:00	Session 1: (Short talks, posters demos, videos)
12:00 - 13:30	Lunch
13:30 - 15:30	Session 2: Workshop Activity: SAR decision-making scenarios & discussion
15:30 - 17:00	Session 3: Panel

Expected Outcomes.

- Build a community across the fields of HRI and HCI who are currently working in or are interested in the area of SAR decision-making.
- Establish challenges around how SARs make decisions, how and in what way these decisions are communicated to their users, and the surrounding ethical implications.
- Derive concrete strategies and practices to address these challenges.

- Disseminate the wider results and outcomes from this workshop through a publication and follow-up workshop at the new National Robotarium in Edinburgh.

9 ACCESSIBILITY

To ensure the inclusion of all participants, we will work with those with accessibility needs and ascertain the support they require through collaboration with the accessibility chairs. At least one workshop organizer will be given the role of Accessibility Chair to ensure all needs are met. Further, we will require all paper submissions to follow the SIGCHI Guide for an Accessible Submission, and all video submissions must contain subtitles/captioning.

10 POST-WORKSHOP PLANS

Authors will be invited to post their accepted workshop paper via arXiv, with links accessible through the workshop website. The organizers will create a publication summarizing the challenges discussed surrounding SAR decision-making and the outcomes of the workshop. A poster version will also be created in order to share this work with the broader community. A mailing list will be created to facilitate future collaboration. Participants will be invited to a subsequent workshop at the National Robotarium in Edinburgh, UK to discover how the workshop outcomes have been integrated into HRI research.

11 CALL FOR PARTICIPATION

Socially Assistive Robots as Decision Makers: Transparency, Motivations, and Intentions is a one-day hybrid workshop which aims to discuss challenges, current practices, and ethical implications of Socially Assistive Robot (SAR) decision-making. We welcome 4-page position or research contribution papers which address challenges within the theme of SARs that make decisions. Some example challenges are listed below. The papers should be in the CHI extended abstract format and be submitted through EasyChair by 11:59 PM on February 20th, 2023. Papers will be reviewed according to suitability to the workshop to contribute to an interesting set of discussions by the participants. Papers will be selected to present a poster or short talk on the day of the workshop, and accepted papers will be invited to submit to arXiv. At least one author of each accepted submission must attend the workshop. More information can be found on our website: <https://sites.google.com/view/sar-decision-making>

- **Which decisions** should be made by the SAR and **who decides this?**
- How should SARs **transparently communicate when reasoning is based on complex processes?**
- How do SARs smoothly integrate into existing relationships in assistive scenarios to **build trust** with users?
- What is the **role of robot embodiment** in communicating explanations?
- How do SARs **recover from unintended/incorrect actions** through communication with the user?
- How should SARs approaches be **adapted for specific populations?**
- Should SARs that make autonomous decisions also have an **explicit moral reasoning framework?**

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